

ABHINAV TARIGOPPULA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [LeetCode](#)

PROFESSIONAL SUMMARY

Machine Learning Engineer building and deploying production AI systems — LLM pipelines, RAG architectures, and real-time model serving and inference optimization. Delivered measurable production impact: **+12% model accuracy** and **-18% inference latency**. Strong in Python, PyTorch, LangChain, and FastAPI with end-to-end ownership from data pipelines to deployment. Active open-source contributor — **8 PRs merged** into major AI frameworks (HuggingFace, agno, mem0, pydantic, marimo) and author of CodeHound, a Python static-analysis tool.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C++

ML & GenAI: PyTorch, TensorFlow, Scikit-learn, LangChain, OpenAI API, Hugging Face Transformers, NLP, RAG, LLM Fine-Tuning, Generative AI, Prompt Engineering

Backend & APIs: FastAPI, Node.js, REST APIs, Vector Databases (FAISS, Pinecone), asyncio, WebSockets

Frontend & DevOps: React.js, Next.js, AWS, Docker, GitHub Actions, CI/CD, pytest, Git, Linux

CS Foundations: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, AST / Static Analysis

WORK EXPERIENCE

Machine Learning & AI Intern | OneStop AI — Remote

May 2025 – Aug 2025

- Improved classification accuracy **12% (78%→90%)** across 3 production models via hyperparameter tuning and architecture search.
- Cut real-time inference latency **18% (220ms→180ms)** by refactoring the preprocessing pipeline and applying model quantization.
- Engineered scalable end-to-end data pipelines (preprocessing, feature engineering, batch processing) integrated with CI/CD for continuous model delivery.
- Strengthened deployment reliability with monitoring and performance-tracking pipelines, ensuring inference stability under production load.

PwC Launchpad Advisory Program (PwC India) | AI & Data Advisory Training

Feb 2026 – Present

- Selected for PwC's flagship GenAI, Prompt Engineering, and Data Systems program; achieved **Level 10 | 1500+ XP** (top performance in cohort).
- Built and optimized LLM prompt workflows and explored enterprise data architectures for real-world advisory use cases.

PROJECTS

PathForge — AI Placement Preparation Platform

[pathforge.online](#)

Python, FastAPI, LangChain, OpenAI API, FAISS/Pinecone, React

- Architected an end-to-end AI placement-prep platform using LLM pipelines for role-specific study material and mock interviews; deployed live.
- Implemented RAG with FAISS/Pinecone for context-aware Q&A and engineered a skill-gap roadmap engine that auto-generates structured learning paths from user profile analysis.
- Applied systematic prompt engineering to improve response accuracy **~30%** and reduce per-query token consumption.

CodeHound — AST-Based Python Static Analyzer

[github.com/kratos0718/codehound](#)

Python, ast, pytest, GitHub Actions, CLI

- Built a zero-dependency static-analysis engine (~750 LOC, CI on Python 3.9/3.11/3.12) that parses code into ASTs to detect async-safety and correctness bugs — blocking calls in async functions, fire-and-forget tasks, mutable defaults, and resource leaks.
- Engineered 6 extensible, independently-tested checks; ran it against agno and OpenAI's SDK to surface previously-unreported bugs — **one merged upstream into agno (25k★)**.
- Packaged as an installable CLI with JSON/CSV output and CI-friendly exit codes for drop-in pipeline integration.

OPEN SOURCE CONTRIBUTIONS

- 8 PRs merged across 18+ organizations** — HuggingFace, agno (25k★), mem0 (35k★), pydantic (22k★), and marimo (11k★, YC) — including async concurrency & correctness bug fixes, a mutable-default (B006) fix, and a new public-API feature; reviewed and merged by core maintainers.
- Shipped a fix into **huggingface_hub v1.17.0** — live on PyPI for every developer who installs the library. Open PRs under review at **OpenAI (Agents SDK)** and **litellm**; bugs independently validated by automated reviewers.

ACHIEVEMENTS

- GitHub Pull Shark** — earned for multiple merged pull requests into major open-source projects.
- Solved **200+ Data Structures & Algorithms problems** (LeetCode) — consistent practice across Easy, Medium, and Hard.
- Authored open-source technical learning resources (async Python, static analysis, and a 500-question core-CS interview guide) on GitHub.
- Participant, **Smart India Hackathon** — India's largest national hackathon (1M+ student participants).

EDUCATION

GITAM University, Andhra Pradesh — B.Tech, Computer Science & Engineering (AI/ML Specialization)

Aug 2023 – 2027

CGPA: 8.12 | Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

FIITJEE Junior College — Intermediate (MPC)